

Machine Learning for My Brother

Lesson 1: Predicting Car Prices Based on Mileage

In this lesson, you will build your first Machine Learning model using Python.

Goal:

Predict the price of a car based on its mileage using Linear Regression.

Required Libraries:

- pandas
- matplotlib
- scikit-learn

Code Example:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

data = {
    'Mileage': [10, 20, 30, 40, 50, 60, 70],
    'Price': [500000, 450000, 400000, 350000, 300000, 250000, 200000]
}
df = pd.DataFrame(data)
X = df[['Mileage']]
y = df['Price']

model = LinearRegression()
model.fit(X, y)

predicted_price = model.predict([[35]])
```

Machine Learning for My Brother

```
print(f"Predicted Price for 35,000 km: R{predicted_price[0]:.2f}")
```

```
plt.scatter(df['Mileage'], df['Price'], color='blue')  
plt.plot(df['Mileage'], model.predict(X), color='red')  
plt.xlabel('Mileage (in thousands)')  
plt.ylabel('Price (Rands)')  
plt.title('Mileage vs Car Price')  
plt.grid(True)  
plt.show()
```

What You Learn:

- How to train a model with data
- How to make predictions
- How to visualize results

This is the beginning of your AI journey.